A divisor of the gcd-product: a proof of Bala's conjecture on OEIS A129364

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Abstract

Let $d(n) = \prod_{k=1}^{n}\prod_{e|k} e^{k/e}$, OEIS A129364, and let $G(n) = \prod_{j,k\le n}\gcd(j,k)$, OEIS A092287. In April 2007 P. Bala conjectured that $d(n)$ divides $G(n)$ for every n. We prove it, in the sharp form

$$v_p\big(G(n)\big) - v_p\big(d(n)\big) = \sum_{i\ge 1} B\big(\lfloor n/p^i\rfloor\big), \qquad B(M) = \sum_{k=1}^{M}\big(M \bmod k\big) \quad \text{(A004125)},$$

for every prime p. Every term on the right is a sum of remainders, hence nonnegative, and the divisibility follows at once. Three ingredients: the layer-cake count for $v_p(G(n))$, Legendre's formula together with the nested-floor identity for $v_p(d(n))$, and the observation that $M^2 - \sum_{k\le M} k\lfloor M/k\rfloor$ is exactly $\sum_{k\le M}(M \bmod k)$.

1 The sequences and the conjecture

OEIS A129364 (Peter Bala, 11 April 2007) has offset 1 and is

$$d(n) = \prod_{k=1}^{n} \text{A066841}(k) = \prod_{k=1}^{n}\prod_{e|k} e^{k/e} = 1, 2, 6, 96, 480, 207360, 1451520, 2972712960, \ldots$$

The entry also records the equivalent closed form

$$d(n) = \prod_{k=1}^{n}\big(\lfloor n/k\rfloor !\big)^k, \tag{1}$$

which we use throughout; the two agree because writing $k = e\cdot m$ turns $\prod_{k\le n}\prod_{e|k} e^{k/e}$ into $\prod_{e\le n} e^{\sum_{m\le n/e} m}$, and the same rearrangement applied to (1) gives the same exponent of each e.

Write $G(n) = \text{A092287}(n) = \prod_{j=1}^{n}\prod_{k=1}^{n}\gcd(j,k)$. The entry carries

> **Conjecture:** $a(n)$ divides A092287(n) for all n — see comments in A129365. — Peter Bala, Apr 2007

As of the "Last modified" line on the live entry (24 August 2026) this is still recorded as an unproven conjecture, with no proof in the entry's links or programs.

Throughout p is a prime, v_p is the p-adic valuation, and $B(M) = \sum_{k=1}^{M}(M \bmod k)$ is A004125, with $B(0) = 0$.

2 Three ingredients

Lemma 1 (layer cake)**.** $v_p\big(G(n)\big) = \sum_{i\ge 1} \lfloor n/p^i\rfloor^2$.

Proof. $v_p(\gcd(j,k)) = \min(v_p(j), v_p(k))$ is the number of $i \ge 1$ with $p^i \mid j$ and $p^i \mid k$. Summing over $j, k \le n$ and exchanging the order of summation, the two divisibility conditions are independent, so level i contributes $\big(\#\{j \le n : p^i \mid j\}\big)^2 = \lfloor n/p^i\rfloor^2$. (This is the formula of A092287, confirmed on that entry by C. R. Greathouse IV in April 2013.) $\square$

Lemma 2 (nested floors)**.** *$\lfloor n/(kq)\rfloor = \lfloor \lfloor n/q\rfloor/k\rfloor$ for positive integers n, k, q.*

Proof. For real x and a positive integer k, $\lfloor x/k\rfloor = \lfloor \lfloor x\rfloor/k\rfloor$; apply with $x = n/q$. $\square$

Lemma 3 (remainder identity)**.** *$M^2 - \sum_{k=1}^{M} k\lfloor M/k\rfloor = B(M)$ for every integer $M \ge 0$.*

Proof. $M^2 = \sum_{k=1}^{M} M$, and $M - k\lfloor M/k\rfloor = M \bmod k$ termwise. $\square$

3 The theorem

Theorem 4. *For every $n \ge 1$ and every prime p,*

$$v_p\big(G(n)\big) - v_p\big(d(n)\big) = \sum_{i\ge 1} B\big(\lfloor n/p^i\rfloor\big).$$

Proof. Put $M_i = \lfloor n/p^i\rfloor$. By (1), Legendre's formula and Lemma 2,

$$v_p\big(d(n)\big) = \sum_{k=1}^{n} k\, v_p\big(\lfloor n/k\rfloor!\big) = \sum_{k=1}^{n}\sum_{i\ge 1} k\big\lfloor \lfloor n/k\rfloor/p^i\big\rfloor = \sum_{i\ge 1}\sum_{k=1}^{n} k\big\lfloor n/(kp^i)\big\rfloor = \sum_{i\ge 1}\sum_{k=1}^{M_i} k\lfloor M_i/k\rfloor,$$

where the last step uses Lemma 2 again in the form $\lfloor n/(kp^i)\rfloor = \lfloor M_i/k\rfloor$, the terms with $k > M_i$ vanishing. Subtracting this from Lemma 1 and applying Lemma 3 at each level i,

$$v_p\big(G(n)\big) - v_p\big(d(n)\big) = \sum_{i\ge 1}\Big(M_i^2 - \sum_{k=1}^{M_i} k\lfloor M_i/k\rfloor\Big) = \sum_{i\ge 1} B(M_i). \qquad \square$$

Corollary 5 (Bala's conjecture)**.** *$d(n)$ divides $G(n)$ for every $n \ge 1$.*

Proof. Each $M \bmod k$ is a nonnegative integer, so $B(M) \ge 0$ and hence $v_p(G(n)) \ge v_p(d(n))$ for every prime p by Theorem 4. A positive rational with nonnegative valuation at every prime is an integer. $\square$

Remark (the same statement lives on two entries)**.** The quotient $G(n)/d(n)$ is A129365, whose own comment list opens with "Conjectures: A) $a(n)$ is always an integer" (Bala, 13 April 2007). That is the present conjecture restated, and the two entries have never been cross-annotated as being the same claim — A129364 merely points at "comments in A129365". Theorem 4 settles both at once: the divisibility asked for here, and the integrality of the quotient asked for there are the same inequality between valuations. Whoever records these should record both, and note that one proof settles both.

Remark (honest assessment)**.** Elementary throughout: the layer-cake count, Legendre, and one floor identity. What is proved is the divisibility exactly as A129364 poses it, together with the exact size of the gap; nothing here is deep, and the value of writing it down is that the exact exponent is on neither entry. The natural destination is a comment on the entry.

4 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) reproduces the first 12 published terms of A129364 from the defining product over divisors; (ii) confirms that the entry's two formulas agree, $\prod_{k\leq n} \mathrm{A066841}(k) = \prod_{k\leq n}(\lfloor n/k\rfloor!)^k$, for $1 \leq n \leq 40$; (iii) confirms the conjecture directly, that $d(n)$ divides $G(n)$, for $1 \leq n \leq 40$; (iv) confirms Theorem 4 for every $1 \leq n \leq 50$ and every prime $p \leq n$, that is 437 pairs (n, p), with 0 mismatches, and prints $B(0), \ldots, B(12) = 0, 0, 0, 1, 1, 4, 3, 8, 8, 12, 13, 22, 17$; and (v) confirms that the quotient $G(n)/d(n)$ reproduces the published DATA of A129365, $1, 1, 1, 1, 1, 2, 2, 2, 6, 48, 48, 48$.

References

[1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A129364`, consulted 24 August 2026; also A066841, A092287, A129365 and A004125.

[2] A. M. Legendre, *Théorie des Nombres*, 3rd ed., Firmin Didot, Paris, 1830.